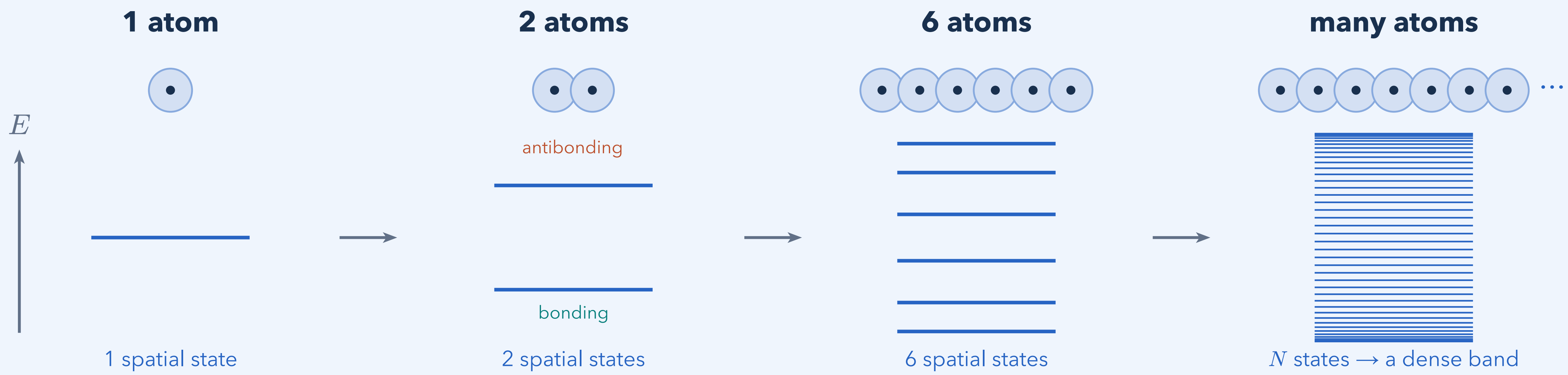
How atoms become energy bands

Atomic orbitals combine into electron waves spread across the solid.

Follow one orbital per atom: coupling creates the energy spread; electron filling decides what the solid can do.

1 MORE ATOMS → MORE ALLOWED LEVELS

Each line is one allowed wave pattern.

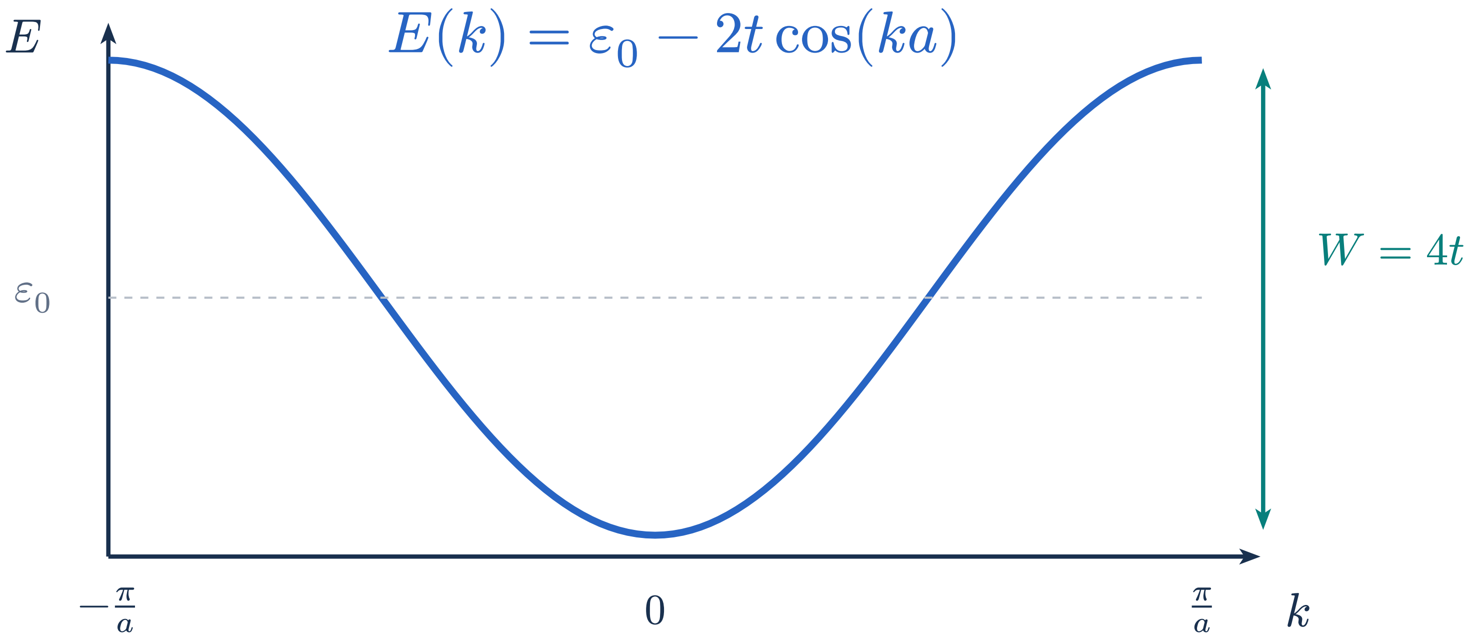


N atomic orbitals → N spatial states → room for $2N$ electrons.

Two opposite spins fit in each spatial state. Pauli exclusion controls occupation; coupling causes the splitting.

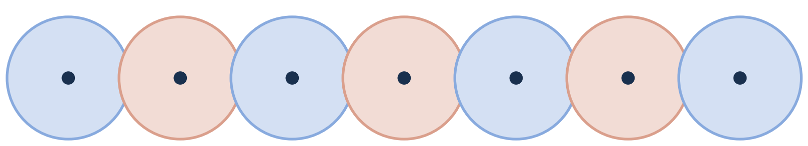
2 THE BAND IS A FAMILY OF WAVES

Minimal model: identical atoms, spacing a, one orbital each, and coupling $-t$ between neighbors ($t > 0$).



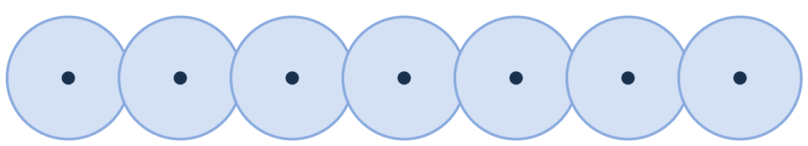
k: phase change per distance; ε_0: isolated orbital energy.
This interval is one Brillouin zone: distinct lattice phases.

HIGH ENERGY • $k = \frac{\pi}{a}$



Alternating phase: antibonding.

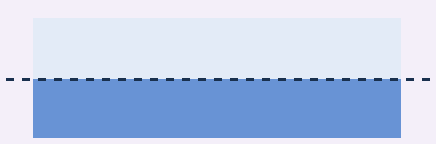
LOW ENERGY • $k = 0$



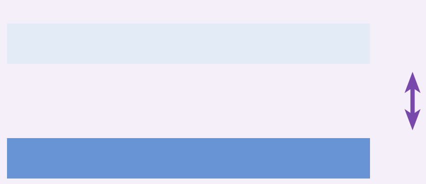
Same phase on neighbors: bonding.

Blue / orange = wavefunction sign, not charge. Each pattern extends across the chain.

3 FILLING MATTERS AS MUCH AS BANDWIDTH



Partly filled band
Nearby empty states permit a metallic response.



Full band + gap above it
A band insulator; a small gap can allow thermally excited carriers.

Dark: occupied; pale: empty. Dashed: Fermi level, the filling boundary at zero temperature. This independent-electron picture can change with overlapping bands, correlations, or symmetry breaking.